Introduction to Object Oriented Programming Techniques

Learing Objectives

After learning this chapter, the students will be able to

- Understand the concept of **OOPS**
- Know the difference between *Procedural*, *Modular* and *Object Oriented Programming*.
- Understand the advantages and disadvantages of Object Oriented Programming.

13.1 Introduction

Object-Oriented Programming (OOP) is the term used to describe a programming approach based on classes and objects. The object-oriented paradigm allows us to organize software as a collection of objects that consist of both data and behaviour. This is in contrast to conventional functional programming practice, that loosely connects data and behaviour.

Since 1980's the word **'object'** has appeared in relation to programming languages, with almost all languages developed since 1990 having object-oriented features. This chapter introduces general OOP concepts.

13.2 Programming Paradigms

Paradigm means organizing principle of a program. It is an approach to programming. There are different approaches available for problem solving using computer. They are Procedural programming, Modular Programming and Object Oriented Programming

13.2.1 Procedural programming

Procedural means a list of instructions were given to the computer to do something. Procedural programming aims more at procedures. This emphasis on doing things.

Important features of procedural programming

- Programs are organized in the form of subroutines or sub programs
- All data items are global
- Suitable for small sized software application
- Difficult to maintain and enhance the program code as any change in data type needs to be propagated to all subroutines that use the same data type. This is time consuming.
- Example: **FORTRAN** and **COBOL**.

11th Std Computer Science_EM Chapter 13.indd 228 06-01-2023 15:11:21